

Resistive Load

Topology	V_{L-DC}	I_{L-DC}	V_{L-RMS}	I_{L-RMS}	I_{D-RMS}	I_{D-DC}	FF (figure factor)	RF(Rip ple factor)	$I_{FRM} = I_{max}$	$V_{RRM} = V_{PIV}$	f_{rp}
1p Half Wave	$0.318V_{m(p)}$	$\frac{0.318V_{m(p)}}{R_L}$	$0.5V_{m(p)}$	$\frac{0.5V_{m(p)}}{R_L}$	$\frac{0.5V_{m(p)}}{R_L}$	$\frac{I_{ave}}{1} = \frac{0.318V_{m(p)}}{R_L}$	1.57	1.21	$\frac{V_{m(p)}}{R_L}$	$V_{s(peak)}$	f_s
1p Full- Wave (center- tapped)	$0.636V_{m(p)}$	$\frac{0.636V_{m(p)}}{R_L}$	$0.707V_{m(p)}$	$\frac{0.707V_{m(p)}}{R_L}$	$\frac{0.5V_{m(p)}}{R_L}$	$\frac{I_{ave}}{2} = \frac{0.318V_{m(p)}}{R_L}$	1.11	0.482	$\frac{V_{m(p)}}{R_L}$	$V_{s(peak)} - V_d$	$2f_s$
1p Full- Wave bridge	$0.636V_{m(p)}$	$\frac{0.636V_{m(p)}}{R_L}$	$0.707V_{m(p)}$	$\frac{0.707V_{m(p)}}{R_L}$	$\frac{0.5V_{m(p)}}{R_L}$	$\frac{I_{ave}}{2} = \frac{0.318V_{m(p)}}{R_L}$	1.11	0.482	$\frac{V_{m(p)}}{R_L}$	$V_{s(peak)} - V_d$	$2f_s$
3P-3pulse (Half Wave)	$0.827V_{m(p)}$	$\frac{0.827V_{m(p)}}{R_L}$	$0.84V_{m(p)}$	$\frac{0.84V_{m(p)}}{R_L}$	$\frac{0.485V_{m(p)}}{R_L}$	$\frac{I_{ave}}{3} = \frac{0.827V_{m(p)}}{3R_L}$	1.0165	0.182	$\frac{V_{m(p)}}{R_L}$	$\sqrt{3}V_{m(p)}$	$3f_s$
3P-6 pulse Full-Wave bridge	$1.654V_{m(p)}$	$\frac{1.654V_{m(p)}}{R_L}$	$1.655V_{m(p)}$	$\frac{1.655V_{m(p)}}{R_L}$	$\frac{0.955V_{m(p)}}{R_L}$	$\frac{I_{ave}}{3} = \frac{0.551V_{m(p)}}{R_L}$	1.0009	0.042	$\frac{\sqrt{3}V_{m(p)}}{R_L}$	$\sqrt{3}V_{m(p)}$	$6f_s$
3P- 12pulse	$3.31V_{m(p)}$	$\frac{3.31V_{m(p)}}{R_L}$	$3.31V_{m(p)}$	$\frac{3.31V_{m(p)}}{R_L}$	$\frac{1.91V_{m(p)}}{R_L}$	$\frac{I_{ave}}{3} = \frac{1.1V_{m(p)}}{R_L}$	1	0.01	$\frac{3.346V_{m(p)}}{R_L}$	$\sqrt{3}V_{m(p)}$	$12f_s$

LM 317

$$V_L = R_P \times \frac{1.25V}{R_{adj}} + 1.25 = 1.25\left(1 + \frac{R_P}{R_{adj}}\right)$$

$$\eta = \frac{P_{out}}{P_{in}} \times 100\%$$

Controlled rectifiers

Three phase half wave controlled rectifier

Resistive load

$$V_{DC} = 0.827V_{m(p)} \cos(\alpha) \quad \text{when } \alpha \leq 30^\circ$$

$$V_{DC} = \frac{0.827}{\sqrt{3}} V_{m(p)} [1 + \cos(\alpha + 30^\circ)] \quad \text{when } \alpha > 30^\circ$$

$$I_{DC} = \frac{V_{DC}}{R_L}$$

Inductive load

$$V_{DC} = 0.827V_{m(p)} \cos(\alpha)$$

$$I_{DC} = \frac{V_{DC}}{R_L}$$

Three phase Full wave controlled rectifier

Resistive load

$$V_{DC} = 1.654V_{m(p)} \cos(\alpha) \quad \text{when } \alpha \leq 60^\circ$$

$$V_{DC} = 1.654V_{m(p)} [1 + \cos(\alpha + 60^\circ)] \quad \text{when } \alpha > 60^\circ$$

$$I_{DC} = \frac{V_{DC}}{R_L}$$

Inductive load

$$V_{DC} = 1.654V_{m(p)} \cos(\alpha)$$

$$I_{DC} = \frac{V_{DC}}{R_L}$$

Conduction Loss

OFF transistor

$$P_{OFF-loss(ave)} = I_{leakage} \times V_{CE(max)} \times \frac{T_{off}}{T}$$

ON transistor

$$P_{ON-loss(ave)} = V_{CE(Sat)} \times I_{C(max)} \times \frac{T_{on}}{T}$$

Switching Loss

Transition ON to OFF state

$$P_{SW(OFF-ave)} = 0.5V_{CE(max)} \times 0.5I_{C(max)} \times \frac{T_{SW(OFF)}}{T} \times 0.636$$

Transition OFF to ON state

$$P_{SW(ON-ave)} = 0.5V_{CE(max)} \times 0.5I_{C(max)} \times \frac{T_{SW(ON)}}{T} \times 0.636$$

Inverter

Full Bridge switching ON time

$$T_{ON} = \frac{V(\text{sampling} - \text{ave}) \times T_S}{V_{DC}}$$

$$m_a = \frac{V_{p-ref(m)} \text{ yields}}{V_{\Delta(m)}} V_{o(m)} = m_a \times V_{DC}$$

Half Bridge Switching ON time

$$T_{ON} = \frac{V(\text{sampling} - \text{ave}) \times T_S}{V_{DC}} + \frac{T_S}{2}$$

$$m_a = \frac{V_{p-ref(m)} \text{ yields}}{V_{\Delta(m)}} V_{o(m)} = m_a \times \frac{V_{DC}}{2}$$